

Pumping Theorem (for CFL)

Let L be a CFL. There exists an integer $p > 0$ such that any string in L with length at least p can be divided into five pieces $w = uvxyz$ satisfying the following conditions.

(1) $uv^ixy^iz \in L$ for any $i \geq 0$,

(2) $|v| + |y| > 0$, and

(3) $|vxy| \leq p$.

Example

Prove that $L = \{a^i b^i c^i : i \geq 0\}$ is not context-free.

Proof.

Assume that L is context-free.

Let p be the pumping length.

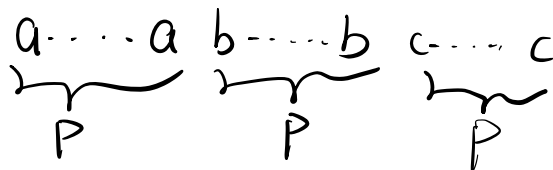
Pick $w = a^p b^p c^p \in L$.

By pumping theorem, w can be written as $w = uvxyz$

(1) $uv^ixy^iz \in L$ for any $i \geq 0$

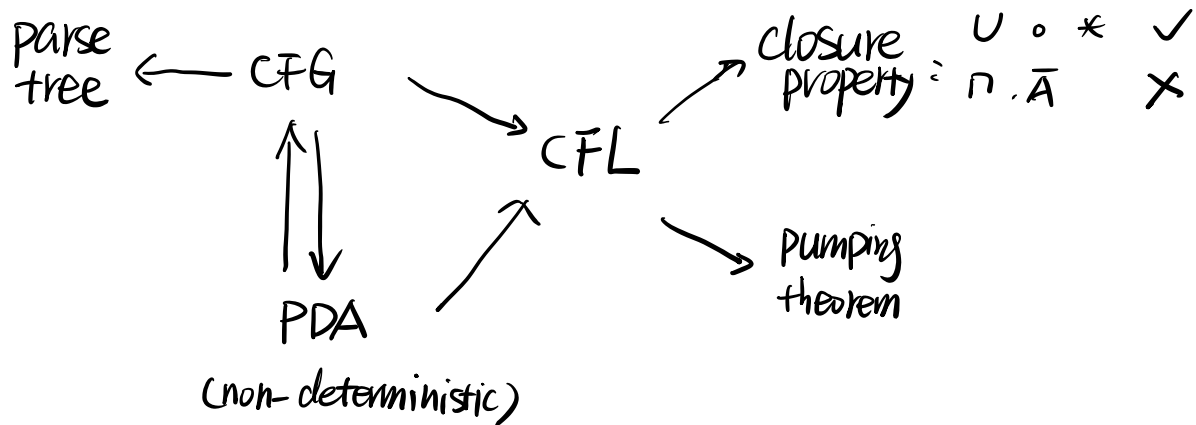
(2) $|v| + |y| > 0$

(3) $|vxy| \leq p$



(3) $\Rightarrow$ at least one of a, b, c does not appear in uvx
+ (2)

$\Rightarrow uv^0xy^0z \notin L$, contradicting with (1).



DPDA

Fact

$\{ww^R : w \in \{a,b\}^*\}$ is not accepted by any DPDA.

Turing Machine

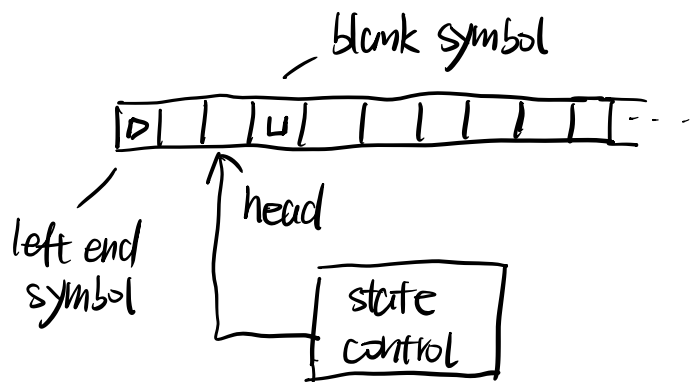
finite automata



PDA



Turing Machine



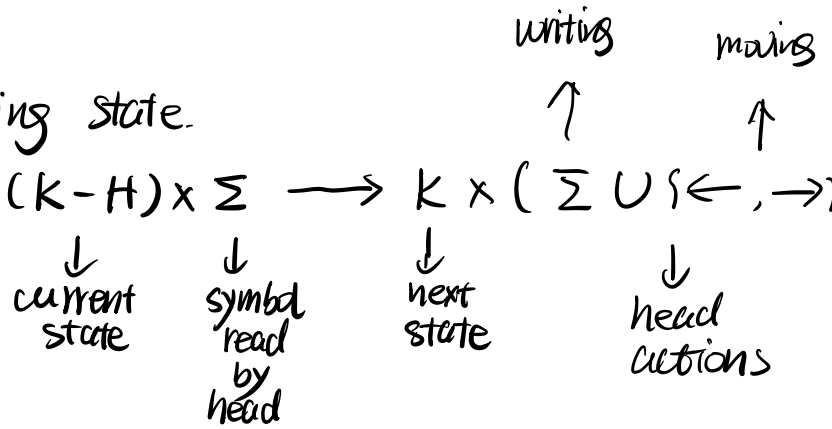
1. one-way infinite tape.

2. its head can move left and right
3. can read and write

Definition

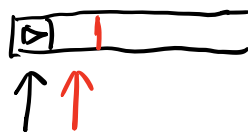
A Turing Machine is 5-tuple $(K, \Sigma, \delta, s, H)$ where

- K is a finite set of states
- Σ is an alphabet containing $\triangleright$ and $\sqcup$
- $s \in K$: initial state
- $H \subseteq K$: a set of halting state.
- transition function $\delta: (K - H) \times \Sigma \longrightarrow K \times (\Sigma \cup \{\leftarrow, \rightarrow\})$



such that

(a) for each $q \in K - H$, $\delta(q, \triangleright) = (p, \rightarrow)$ for some $p \in K$.



(b) for any $q \in K - H$, any $a \in \Sigma$
 if $\delta(q, a) = (p, b)$, then $b \neq \triangleright$
 never write $\triangleright$

A configuration is a member of

$$K \times \triangleright (\Sigma - \{\triangleright\})^* \times (\underbrace{(\Sigma - \{\triangleright, \sqcup\})^* (\Sigma - \{\triangleright, \sqcup\})}_{\text{current}} \cup \{\epsilon\})$$

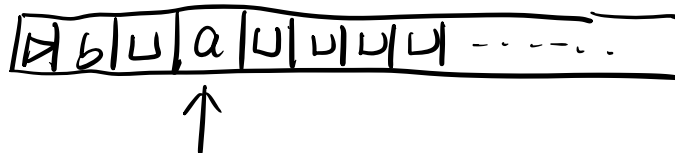
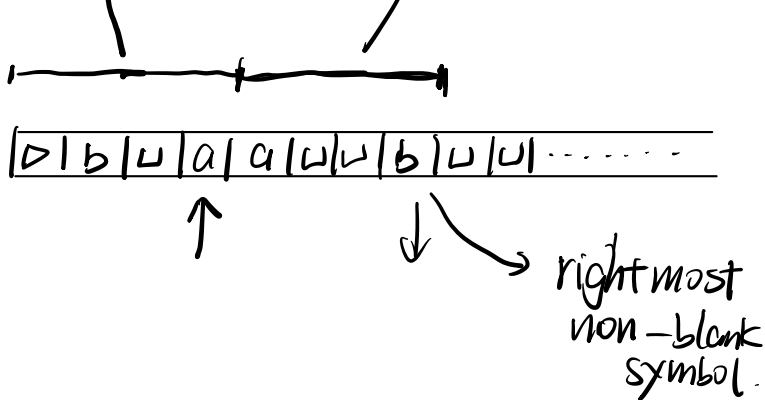
current state

$(q, \triangleright b \underline{u} a, a \sqcup \sqcup b)$

$(q, \triangleright b \sqcup \underline{a} a \sqcup \sqcup b)$

$(q, \triangleright b \underline{u} a, e)$

$(q, \triangleright b \underline{u} a)$

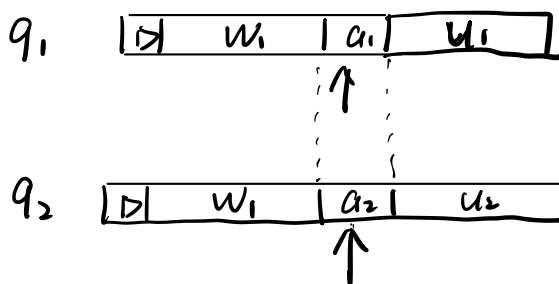


→ halting configuration if $q \in H$.

We $(q_1, \triangleright w_1 \underline{a_1} u_1) \vdash_m (q_2, \triangleright u_2 \underline{a_2} u_2)$ if

(1) $w_1 = w_2$, $u_1 = u_2$, $a_2 \in \Sigma - \{\triangleright\}$, and $\delta(q_1, a_1) = (q_2, a_2)$

writing

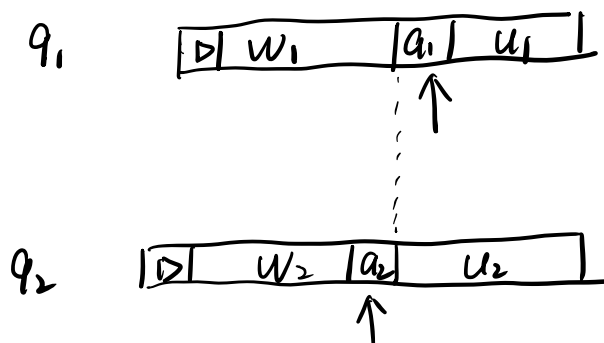


(2) $\delta(q_1, a_1) = (q_2, \leftarrow)$, $w_1 = w_2 a_2$, and

(a) $u_2 = e$, if $u_1 = e$ and $a_1 = \sqcup$

(b) $u_2 = a_1 u_1$, otherwise.

move left

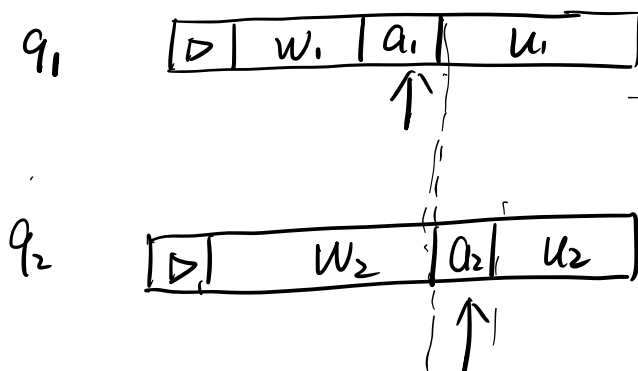


(3) $\delta(q_1, a_1) = (q_2, \rightarrow)$, $w_2 = w_1 a_1$, and

(a) $u_1 = e$ if $u_2 = e$ and $a_2 = \sqcup$

(b) $u_1 = \underline{a_2 u_2}$, otherwise.

$u_1 = \sqcup$



t_M^*

We say two configuration $C t_M^* C'$ if

(i) $C = C'$, or

(ii) there exists $C_0, C_1, \dots, C_k$ for some $k \geq 1$. st

$$C = C_0 t_M C_1 \dots t_M C_k = C'$$

M halts whenever it reaches a halting configuration.

Example.

$$M = (K, \Sigma, \delta, s, H)$$

$$K = \{s, h\}$$

$$\Sigma = \{a, b\}$$

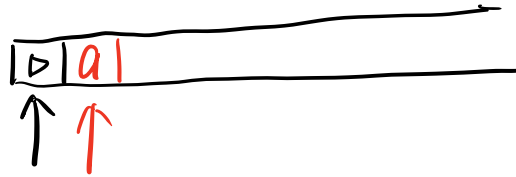
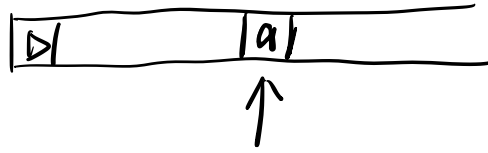
$$H = \{h\}$$

(q, a)	$\delta(q, a)$
$(s, \triangleright)$	$(s, \rightarrow)$
(s, a)	$(s, \rightarrow)$
$(s, \sqcup)$	$(s, \rightarrow)$
(s, b)	(h, b)

Exercise.

Fix some alphabet Σ containing $\triangleright$ and $\sqcup$.

1) for any $a \in \Sigma - \{\triangleright\}$, symbol writing machine M_a

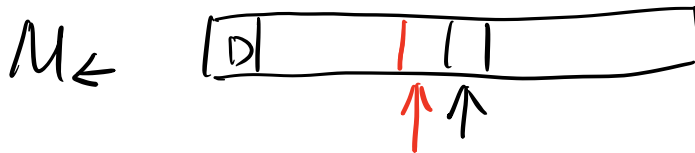


$$M_a = (\{s, h\}, \Sigma, \delta, s, \{h\})$$

$$\delta(s, b) = (h, a) \text{ for each } b \in \Sigma - \{\triangleright\}.$$

$$\delta(s, \triangleright) = (s, \rightarrow)$$

2) head-moving machines $M_{\leftarrow}$ and $M_{\rightarrow}$



$$M_{\leftarrow} = (\{s, h\}, \Sigma, \delta, s, \{h\})$$

$$\delta(s, b) = (h, \leftarrow) \text{ for each } b \in \Sigma - \{\triangleright\}$$

$$\delta(s, \triangleright) = (h, \rightarrow).$$

$M_{\rightarrow}$

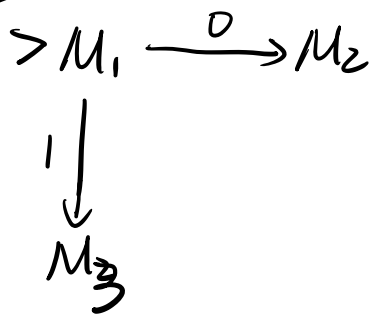
13) left-shifting machine $S \leftarrow$

for any string $w \in (\Sigma - \{0, 1\})^*$,

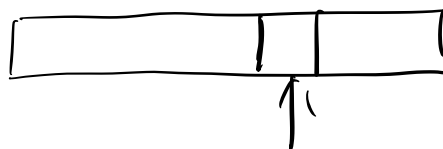
$$\triangleright \square \square \square w \square \rightarrow \triangleright \square w \square$$

basic machines $M_a, M_{\leftarrow}, M_{\rightarrow}$

Combining machines



$M.$



1. run M_1 until it would halt
2. if the current symbol is 0
run M_2
3. elseif the current symbol is 1
run M_3
4. else // the current symbol is not 0 or 1.
halt.

$$M_i = (K_i, \Sigma_i, \delta_i, S_i, H_i) \text{ for } i=1, 2, 3 \quad \Sigma_1 = \Sigma_2 = \Sigma_3 = \Sigma$$

$$M = (K, \Sigma, \delta, s, H)$$

- $K = K_1 \cup K_2 \cup K_3 \cup H$
- $s = s_1$
- $H = H_2 \cup H_3 \cup H_1$
- for $q \in K - H$, for each $a \in \Sigma$

(i) If $q \in K_1 - H_1$

$$\delta(q, a) = \delta_1(q, a)$$

(ii) If $q \in H_1$,

$$\delta(q, 0) = (s_2, 0)$$

$$\delta(q, 1) = (s_3, 1)$$

$$\delta(q, a) = (h, \rightarrow) \text{ for any } a \in \Sigma - \{0, 1\}$$

(iii) If $q \in K_2 - H_2$

$$\delta(q, a) = \delta_2(q, a)$$

(iv) If $q \in K_3 - H_3$

$$\delta(q, a) = \delta_3(q, a)$$

Example

$$> R \xrightarrow{0, 1, \square, \sqcup} R$$

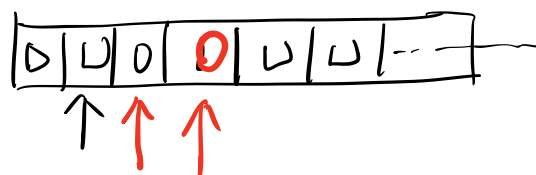
$$> R \xrightarrow{a \neq \sqcup} R_a$$

$$\text{If } \Sigma = \{0, 1, \square, \sqcup\}$$

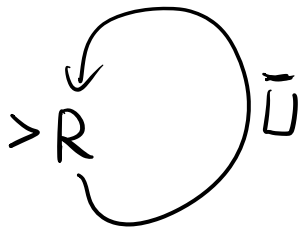
$$> R \longrightarrow R$$

$$> RR$$

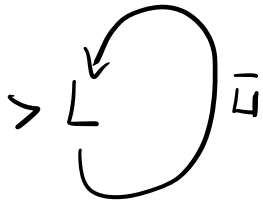
$$> R^2$$



$\bar{U}$: any symbol other than U



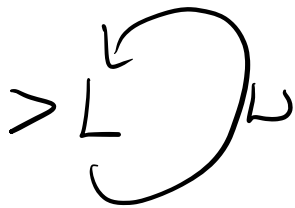
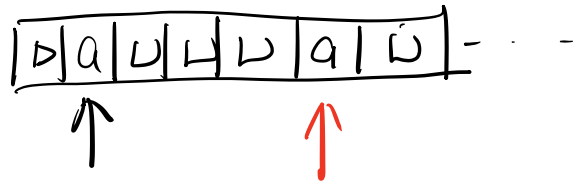
R_U : find the first U to the right of head.



L_U :

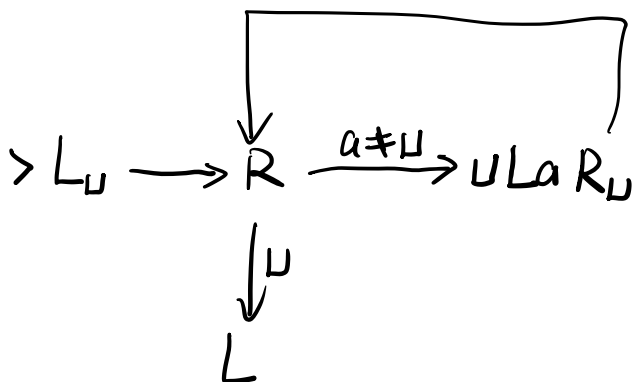
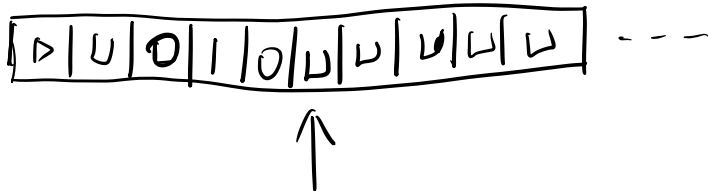


$R_{\bar{U}}$



$L_{\bar{U}}$

Left-shifting machine $S \leftarrow \triangleright U U W U \rightarrow \triangleright U W U$



Copy machine

DUWU $\longrightarrow$ DUWUWU

